MILITARY SPECIFICATION

COATING COMPOUND, THERMAL INSULATION, THREE-COMPONENT, INTUMESCENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the requirements for three component sprayable, thermal insulating compounds for coating the exterior surfaces of aircraft ordnance to provide protection from high-flux thermal environments.

1.2 Classification. The coating compound shall be furnished in the following classes as specified (see 6.2):

Class 1 - Maximum flexibility properties (see 6.6)
Class 2 - Maximum tensile strength properties (see 6.6)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

TT-M-261	Methyl Ethyl Ketone, Technical
TT-M-268	Methyl Isobutyl Ketone (For Use in Organic Coatings)
TT-P-143	Paint, Varnish, Lacquer, and Related Materials, General Specification for Packaging, Packing and Marking of

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:38Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS

Federal (Continued)

TT-T-548	Toluene, Technical
QQ-A-250/5	Aluminum Alloy, Alclad 2024, Plate and Sheet
QQ-A-250/11	Aluminum Alloy 6061, Plate and Sheet
RR-S-366	Sieve, Standard for Testing Purposes
PPP-D-729	Drums, Metal, 55 Gallon (For Shipment of Non-corrosive Materials)
PPP-P-704	Pail, Shipping, Steel (1 Through 12 Gallon)

Military

MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-A-8625	Anodic Coatings, For Aluminum and Aluminum Alloys
MIL-P-23377	Primer Coating; Epoxy-Polyamide, Chemical and Solvent Resistant
MIL-C-81706	Chemical Conversion Materials For Coating Aluminum and Aluminum Alloys
MIL-C-81773	Coating, Polyurethane, Aliphatic, Weather Resistant

STANDARDS

Federal

Fed. Test Method Std. No. 141	Paint, Varnish, Lacquer and Related Materials, Methods of Inspection, Sampling and Testing
Fed. Test Method Std. No. 406	Plastics: Methods of Testing

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-810	Environmental Test Methods

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D 56	Test for Flash Point by Tag Closed Tester
ASTM D 92	Flash and Fire Points by Cleveland Open Cup
ASTM D 445	Viscosity of Transparent and Opaque Liquids (Kinematic and Dynamic Viscosities)
ASTM D 891	Specific Gravity of Industrial Aromatic Hydrocarbons and Related Materials
ASTM D 1002	Strength Properties of Adhesives in Shear by Tension Loading (Metal-to-Metal)
ASTM D 1014	Conducting Exterior Exposure Tests of Paints on Steel
ASTM D 1217	Density and Specific Gravity of Liquids by Bingham Pycnometer
ASTM D 1310	Flash Point of Liquids by Tag Open-Cup Apparatus
ASTM D 1364	Water in Volatile Solvents (Fischer Reagent Titration Method)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

American Society for Testing and Materials (Continued)

ASTM D 1417	Synthetic Rubber Latices (Styrene-Butadiene Copolymers)
ASTM D 1461	Moisture or Volatile Distillates in Bituminous Paving Mixtures
ASTM D 1475	Density of Paint, Varnish, Lacquer, and Related Products
ASTM D 1544	Color of Transparent Liquids (Gardner Color Scale)
ASTM D 1638	Urethane Foam Isocyanate Raw Materials
ASTM D 1652	Epoxy Content of Epoxy Resins
ASTM D 1747	Refractive Index of Viscous Materials
ASTM D 1748	Rust Protection by Metal Preservatives in the Humidity Cabinet
ASTM D 2196	Rheological Properties of Non-Newtonian Materials
ASTM D 2369	Volatile Content of Paints
ASTM E 20	Analysis by Microscopical Methods for Particle-size Distribution of Particulate Substances of Subsieve Sizes
ASTM E 70	pH of Aqueous Solutions with the Glass Electrode

(Application for copies of ASTM Standards should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

National Aeronautics and Space Administration

Specification A-16839(E)	Ammonium Salt of 4-Nitroaniline-2-Sulfonic Acid

(Application for copies of NASA specifications should be addressed to the Chemical Research Projects Office, NASA-Ames Research Center, Moffett Field, California 94035.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 Qualification. The coating compound furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.3 and 6.3). Any change in the formulation of a qualified product will necessitate its requalification. The material supplied under the contract shall be identical, within manufacturing tolerances, to the product receiving qualification.

3.2 Components. The coating compound shall be furnished as a kit consisting of three separately packaged components (see Table I).

3.2.1 Formulation. When the components of Class 1 or 2 are mixed in the following ratios, a coating compound meeting the requirements of this specification shall result. Component III is first added to Component I and stirred thoroughly. Component II shall then be added to the admixture of Components I and III and stirred thoroughly.

COMPONENTS	PARTS BY WEIGHT	
	CLASS 1	CLASS 2
I	841.0	761.7
II	117.5	133.6
III	10.0	39.2

3.3 Composition. The composition of Class 1 and 2 Components shall be as specified in Table I.

3.4 Toxicity. The material shall have no adverse effect on the health of personnel when used under the conditions specified in 5.3.1. Questions pertinent to this effect shall be referred by the procuring activity to the appropriate department medical service who will act as an advisor to the procuring activity.

3.5 Component requirements.

3.5.1 Component I. Component I of Class 1 and Class 2 Compound shall conform to the requirements specified in Table II, 3.5.4.1 and 3.5.5 (see Table XII).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

TABLE I

COMPONENT COMPOSITIONS 1/ 2/

CLASS 1		CLASS 2	
COMPONENT I			
Ingredients	Parts by Weight	Ingredients	Parts by Weight
Polysulfide polymer, liquid	200.0	Polysulfide polymer,	100.0
Ammonium salt of 4-nitroaniline -2- sulfonic acid	550.0	Ammonium salt of 4-nitroaniline -2- sulfonic acid	404.7
Toluene	84.0	Methyl Ethyl Ketone (MEK)	205.6
Silica powder	7.0	Silica Fibers	51.4
COMPONENT II			
Ingredients	Parts by Weight	Ingredients	Parts by Weight
Epoxy resin, liquid	100.0	Epoxy resin, liquid	100.0
Silica powder	5.0	Toluene	33.6
Glass fibers	5.0		
Methyl Isobutyl Ketone (MIBK)	7.5		
COMPONENT III			
Ingredients	Parts by Weight	Ingredients	Parts by Weight
2, 4, 6-Tri (dimethyl-aminomethyl) phenol	10.0	2, 4, 6-Tri (dimethyl-aminomethyl) phenol	19.6
		Toluene	19.6

1/ Ingredients suggested for use in the manufacture of each component are listed in 6.5.

2/ Methods of processing and sequence of adding ingredients is important in securing the specified physical properties; therefore, processing of materials in accordance with compositional requirements of Table I is not a guaranty of acceptability under this specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

TABLE II

COMPONENT I REQUIREMENTS

CHARACTERISTICS	REQUIREMENTS			
	CLASS 1, COMPONENT I		CLASS 2, COMPONENT I	
	MINIMUM	MAXIMUM	MINIMUM	MAXIMUM
Total volatile content percent by weight	9.0	9.5	25.0	27.0
Viscosity, centipoises	50,000	100,000	2500	3200
Weight per gallon, lbs	11.6	11.9	10.5	10.9
Flash Point, ^{o}F (Tag Closed Cup)	25	-	0	-

3.5.2 Component II. Component II of Class 1 and Class 2 compound shall conform to the requirements specified in Table III, 3.5.4.1 and 3.5.5 (see Table XII).

TABLE III

COMPONENT II REQUIREMENTS

CHARACTERISTICS	REQUIREMENTS			
	CLASS 1, COMPONENT II		CLASS 2, COMPONENT II	
	MINIMUM	MAXIMUM	MINIMUM	MAXIMUM
Volatile Content percent by Weight	6.0	9.0	24.0	26.0
Viscosity, Centipoises	1900	2900	80	100
Weight per gallon, lbs	9.4	9.8	8.8	9.2
Flash Point, ^{o}F (Tag Closed Cup)	94	-	40	-

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:38Z
Check the source to verify that this is the current version before use.

3.5.3 Component III.

3.5.3.1 Class 1. Class 1, Component III compound (2, 4, 6-Tri (dimethylaminomethyl) phenol) shall be a clear dark amber liquid free of phase separation and suspended particles and shall conform to the requirements specified in Table IV, 3.5.4.2 and 3.5.5. In addition, the material shall produce the infrared (IR) spectral configuration as indicated in Figure 1 (see Table XII and 4.6.2.1).

TABLE IV

CLASS 1, COMPONENT III REQUIREMENTS
(2, 4, 6-TRI (DIMETHYL-AMINOMETHYL) PHENOL)

CHARACTERISTICS	REQUIREMENTS	
	MINIMUM	MAXIMUM
Amine value	580	640
Weight per gallon, lbs	8.0	8.2
Flash Point, °F (Tag Open Cup)	300	-
Water Content, percent by weight	-	0.06

3.5.3.2 Class 2. Class 2, Component III compound shall conform to the requirements specified in Table V, 3.5.4.1 and 3.5.5 (see Table XII and 4.6.2).

TABLE V

CLASS 2, COMPONENT III REQUIREMENTS

CHARACTERISTICS	REQUIREMENTS	
	MINIMUM	MAXIMUM
Amine value	580	640
Weight per gallon, lbs	7.5	7.7
Flash Point, °F (Tag Closed Cup)	18	-
Water Content, percent by weight	-	0.3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

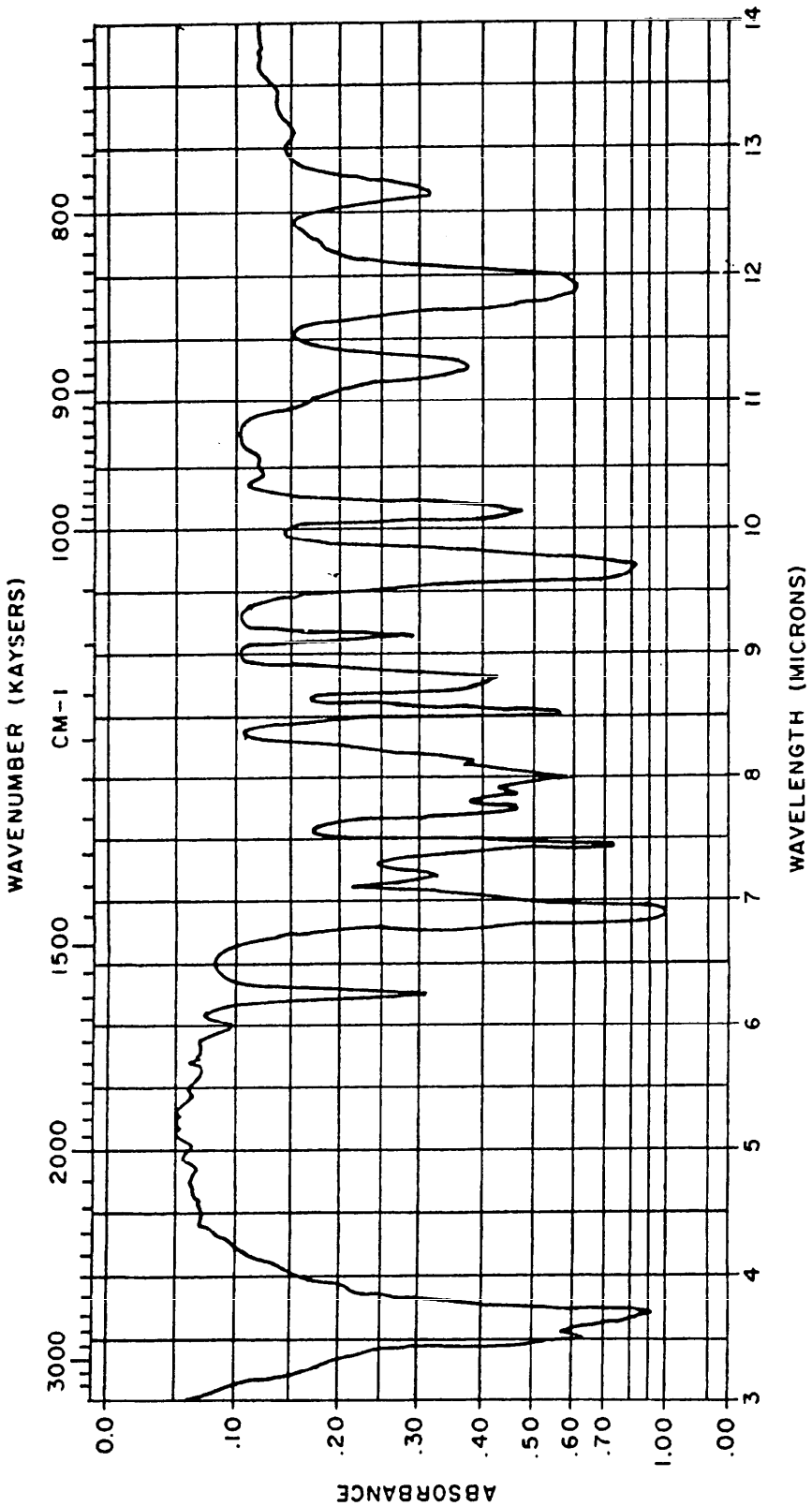


Figure 1. Infrared Absorption Curve, 2, 4, 6-Tri (Dimethyl-aminomethyl) Phenol

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

3.5.4 Condition in container.

3.5.4.1 Components I and II. Components I and II, stored in a closed container, shall be capable of being mixed by hand with a paddle or mechanical means to a smooth homogeneous condition. There shall be no trace of grit, seeds, skins or lumps, and no evidence of settling which cannot be readily dispersed by mixing.

3.5.4.2 Component III. Component III, stored in a closed container, shall show no trace of skin, separation or settling.

3.5.5 Storage stability. The previously unopened packaged Components I, II and III shall meet the respective requirements specified in Tables II through V and the applicable requirements of 3.5.4 for a minimum period of 6 months when stored at 65 $\pm$ 25°F from the date the product is received by the procuring activity (see 4.6.1.2).

3.6 Ingredients.

3.6.1 Liquid polysulfide polymer. The polysulfide compound shall conform to the properties specified in Table VI. In addition, the material shall produce the Infrared (IR) spectral configuration as indicated in Figure 2 (see 4.6.3.1.1 and Table XII).

TABLE VI

LIQUID POLYSULFIDE PROPERTY REQUIREMENTS

PROPERTY	REQUIREMENTS	
	MINIMUM	MAXIMUM
Color (Gardner)	-	12
Moisture content, percent	-	0.1
pH (Water extract)	6.5	7.5
Viscosity at 25°C, cps	940	1440
Specific Gravity at 25°C	1.260	1.283

3.6.2 Ammonium salt of 4-nitroaniline-2-sulfonic acid. The ammonium salt shall conform to the requirements of NASA Specification No. A-16839 (E).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

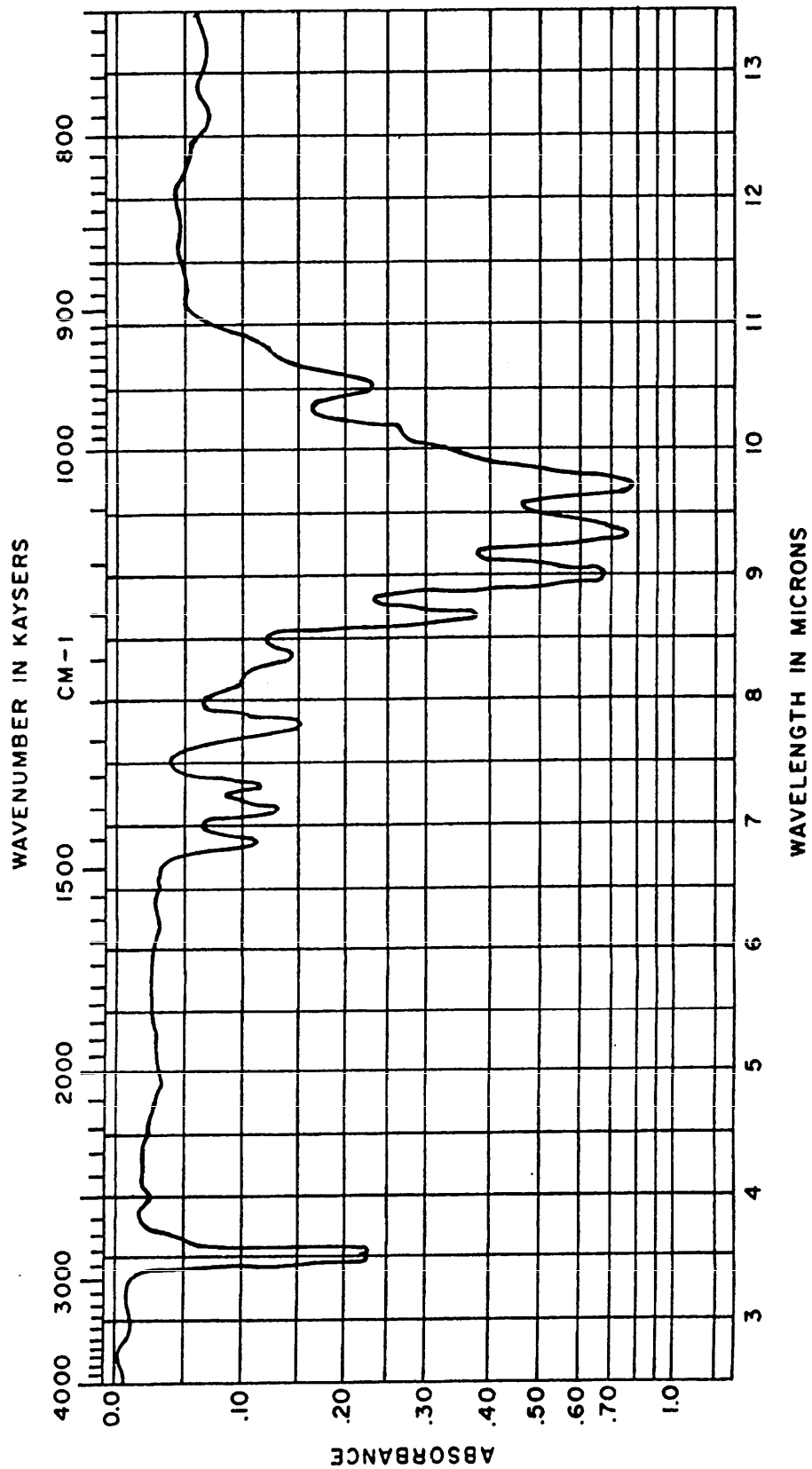


Figure 2. IR Spectrum of Liquid Polysulfide Polymer (As Liquid Between Two Salt Plates)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

3.6.3 Toluene. Toluene shall conform to the requirements of TT-T-548.

3.6.4 Silica powder. The silica powder shall conform to the properties specified in Table VII.

TABLE VII

SILICA POWDER PROPERTY REQUIREMENTS

PROPERTY	REQUIREMENT		TEST PARAGRAPH
	MINIMUM	MAXIMUM	
pH (4 percent aqueous dispersion)	3.5	4.2	4.6.3.4.1
Residue percent, 325 sieve	-	0.02	4.6.3.4.2
Bulk density, lbs/cu. ft.	-	4.4	4.6.3.4.3
Surface area m^2/gm (BET)	175	225	4.6.3.4.4

3.6.5 Liquid epoxy resin. The epoxy resin shall conform to the properties specified in Table VIII. In addition, the material shall produce the Infrared (IR) spectral configuration as indicated in Figure 3 (see 4.6.3.5 and Table XIV).

TABLE VIII

LIQUID EPOXY, RESIN PROPERTY REQUIREMENTS

PROPERTY	REQUIREMENT	
	MINIMUM	MAXIMUM
Epoxide equivalent	185	192
Viscosity at 25°C, poises	100	160
Color (Gardner)	-	4
Refractive Index at 25°C	1.570	1.575
Flash point, °F (Tag open cup)	175	-
Density at 20°C, g/ml	1.163	1.173

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

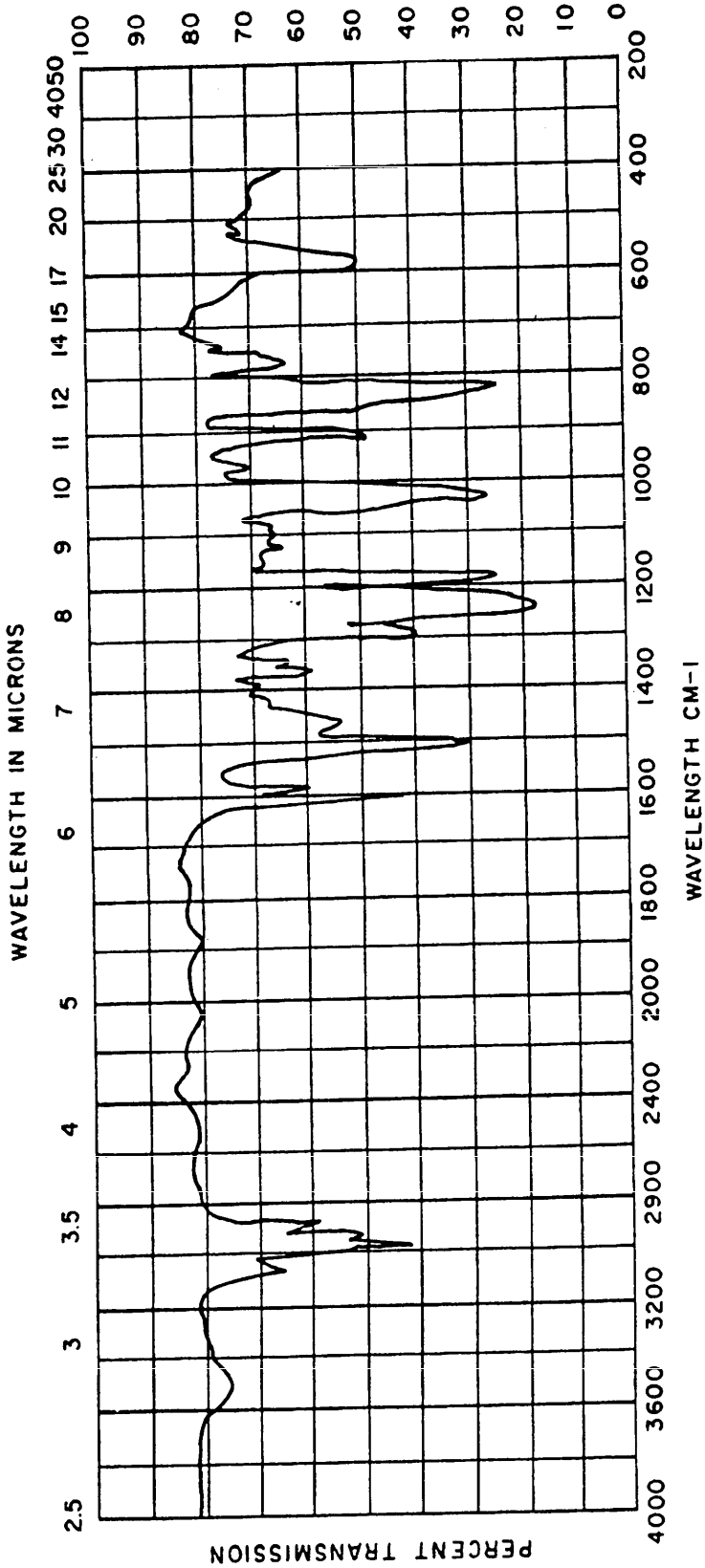


Figure 3. Infrared Absorption Curve of Liquid Epoxy Resin (As Liquid Between Two Salt Plates)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

3.6.6 Glass fibers. The glass fibers shall be of the filament form, powdery, free flowing, gritty to the touch and shall have a cationic sizing so as to be compatible with epoxy resins, and shall conform to the requirements of Table IX.

TABLE IX

GLASS FIBER PROPERTY REQUIREMENTS

PROPERTY	REQUIREMENT		TEST PARAGRAPH
	MINIMUM	MAXIMUM	
Bulk density, gms/cc	0.55	0.90	4.6.3.6.1
Ignition loss, percent	-	0.50	4.6.3.6.2
Fiber length, inches	0.028	0.034	4.6.3.6.3
Fiber diameter, inches	0.00047	0.00055	4.6.3.6.3

3.6.7 Methyl Isobutyl Ketone. Methyl isobutyl ketone shall conform to the requirements of TT-M-268.

3.6.8 Methyl Ethyl Ketone. Methyl ethyl ketone shall conform to the requirements of TT-M-261.

3.6.9 Silica fibers. The silica fibers shall be of the filament form containing no binders or finishes, and shall conform to the requirements of Table X.

TABLE X

SILICA FIBER PROPERTY REQUIREMENTS

PROPERTY	REQUIREMENT		TEST PARAGRAPH
	MINIMUM	MAXIMUM	
Bulk density, gms/cc	0.17	0.25	4.6.3.6.1
Ignition loss, percent	-	1.0	4.6.3.6.2
Fiber length, inches	0.057	0.069	4.6.3.6.3
Fiber diameter, inches	0.0003	0.0005	4.6.3.6.3

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:38Z
Check the source to verify that this is the currentversion before use.

3.6.10 2, 4, 6-Tri (dimethylaminomethyl) phenol. The ingredient 2, 4, 6-Tri (dimethylaminomethyl) phenol shall conform to the requirements of 3.5.3.1.

3.7 Application (Class 1 or 2). The admixed coating compound shall be a freely working product. When applied as specified in 4.6.4, the compound shall spray satisfactorily without running or sagging.

3.8 Thermal efficiency. The thermal efficiency of Class 1 or Class 2 coating system (coating compound plus polyurethane topcoat) shall be a minimum of 1.4 seconds per mil to reach a backface temperature of 260°C (500°F) (see 4.6.5) before and after being subjected to the humidity test of 4.6.11.

3.9 Tensile properties. The minimum tensile properties of the coating compound (see 4.6.6) shall be as follows:

TEMPERATURE (°F)	TENSILE STRENGTH (PSI)		ELONGATION-AT-BREAK (PERCENT)	
	CLASS 1	CLASS 2	CLASS 1	CLASS 2
-40	1500	2700	1.5	0.8
75	160	1400	25.0	0.5
140	130	700	15.0	1.0

3.10 Color. The components shall be supplied in the manufactured color. Tinting agents shall not be used with any component specified in Table I.

3.11 Hardness. The hardness of Class 1 or Class 2 compound, when cured as specified in 4.6.7.2 shall conform to the following:

	MINIMUM SHORE A HARDNESS
Class 1	80
Class 2	90

3.12 Adhesion. Panels coated as specified in 4.6.7.3, when bent around a mandrel 1 ± 0.003 inch in diameter, shall not exhibit adhesive failure from the epoxy primer or polyurethane topcoat. Cracking of the coating is acceptable and shall not be cause for rejection.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

3.13 Resistance properties.

3.13.1 Lubricating oil resistance. The cured coating compound shall withstand immersion in oil for a period of 4 hours at a temperature of 250 $\pm 4^{\circ}$F. Four hours after removal, the coating shall show no blistering, softening or other defects. Discoloration shall not be cause for rejection (see 4.6.8).

3.13.2 Fungus resistance. The cured coating system shall be non-nutrient after subjection to the fungus test specified in 4.6.9.

3.13.3 Impact resistance. The Class 1 cured coating system when tested as specified in 4.6.10 shall require more than 80 inch-pounds on direct (coated surface) impact and 10-inch-pounds minimum on reverse impact before cracking or flaking is evident. The Class 2 cured coating system shall require more than 5 inch-pounds on direct impact before cracking or flaking is evident. Reverse impact shall not be required for Class 2 material.

3.13.4 Humidity resistance. Upon removal of the panels from the cabinet (see 4.6.11), Class 1 cured coating system shall have a minimum Shore A hardness of 65 and Class 2 cured coating system shall have a minimum Shore A hardness of 80.

3.13.5 Resistance to temperature shock. There shall be no evidence of cracking, loss of adhesion, or film deterioration of the coating system when tested as specified in 4.6.12.

3.13.6 Weather resistance. There shall be no evidence of cracking or loss of adhesion of the coating after 1 year of exposure as specified in 4.6.13. After exposure, the thermal efficiency of the coating shall be not less than 1.2 seconds per mil to reach a backface temperature of 260°C (500°F). Subjection of the panels to the humidity test after exposure shall not be required.

3.14 Workmanship. The components shall be processed in a manner that will produce the highest quality material necessary to meet the requirements of this specification. Each component shall be uniform in quality and consistency after stirring, and free of agglomerates or foreign particles. There shall be no other defects present that would render the end product unsuitable for its intended purpose.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection of the coating compound shall be classified as follows:

a. Qualification inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 Qualification.

4.3.1 Inspection. The qualification inspection of the coating compound shall consist of examinations and all the tests listed under 4.6, "Test Methods", and a review for approval of the suppliers report of tests.

4.3.1.1 Retention of qualification. In order to retain qualification of products approved for listing on the Qualified Products List (QPL), the manufacturer shall verify by certification to the qualifying activity that his product(s) comply with the requirements of this specification. Unless otherwise specified by the qualifying activity, the time of periodic verification by certification shall be in two-year intervals from the date of original qualification.

4.3.2 Samples. Qualification test samples shall consist of wet samples, coated test panels, and specimens necessary for performing all the tests specified under 4.6. Wet samples of Components I, II, and III of Class 1 or Class 2 material shall be selected as specified in Method 1031 of Fed. Test Method Std. No. 141 and shall be furnished in one gallon kits so that when admixed in the ratio specified in 3.2.1, a volume of approximately 5 gallons of coating shall be available for test purposes. One quart samples of each ingredient listed in Table I and one quart of polyurethane topcoat conforming to MIL-C-81773, white, Color No. 17875 shall also be furnished. The samples and test specimens shall be forwarded to the Commanding Officer, U. S. Naval Air Development Center, Warminster, Pennsylvania 18974, Attention: Code 30222. (See 6.3 for further information concerning submittal of samples.) The samples shall be legibly marked with the following information:

a. Coating Compound, Thermal Insulation, Three-Component, Sprayable.

b. Class (1 or 2 as applicable)

c. Component (I, II or III as applicable)

d. Specification MIL-C-81947A

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

e. Name and address of manufacturer

f. Location and identity of the plant which produced the samples

g. Manufacturer's identification

h. Submitted by (name) (date) for qualification tests in accordance with MIL-C-81947A under authorization of (reference authorizing letter) (see 6.3).

4.3.2.1 Test samples for qualification shall be submitted with a notarized report of all tests specified under 4.6 "Test Methods", except for the storage stability, and weather resistance tests.

4.4 Quality conformance inspection. Each lot of coating compound. submitted for acceptance under contract after approval of the qualification inspection samples, shall be subjected to quality conformance inspection. Quality conformance inspection shall consist of reports, sampling levels and the examinations specified in 4.4.1 through 4.4.4, and the lot acceptance tests of Table XI.

TABLE XI

LOT ACCEPTANCE TESTS

ITEM	REQUIREMENT PARAGRAPH	TEST METHOD
COMPONENT I	Table II	Table XII
Condition in container	3.5.4.1	4.6.1.1
COMPONENT II	Table III	Table XII
Condition in container	3.5.4.1	4.6.1.1
COMPONENT III	3.5.3 and 4.4.1.1	Table XII
Condition in container	3.5.4.2	4.6.1.1
ADMIXTURE		
Application	3.7	4.6.4
Color	3.10	4.6.7.1
Hardness	3.11	4.6.7.2
Adhesion	3.12	4.6.7.3
Resistance to lubricating oil	3.13.1	4.6.8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

4.4.1 Report of tests. The manufacturer shall submit test reports to the Government representative, in accordance with Method 1031 of Fed. Test Method Std. No. 141, for each lot showing the results including numerical values when applicable of all the tests specified in Table XI. Reports of numerical tests as satisfactory are not acceptable.

4.4.1.1 Ingredient tests. In lieu of performing tests on each ingredient (4.6.3 through 4.6.3.9), and 2, 4, 6-Tri (dimethylaminomethyl) phenol (3.5.3.1), the manufacturer may submit a notarized certification signed by a responsible official of its management, attesting that each ingredient utilized in the coating formulation conforms to the requirements specified herein. The specific test data on each ingredient may be obtained from the ingredient manufacturer for inclusion in the certification.

4.4.2 Sampling.

4.4.2.1 Lot. An inspection lot shall consist of all the coating compound of the same class produced by one manufacturer during a period not to exceed 24 hours, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.4).

4.4.2.2 Sampling for lot acceptance tests. Unless otherwise specified, sampling for lot acceptance tests shall be in accordance with Method 1021 of Fed. Test Method Std. No. 141. The sample shall be sufficient in size to conduct all required tests.

4.4.2.3 Sampling for visual inspection of filled containers. A random sample of filled containers shall be selected in accordance with MIL-STD-105, Inspection Level S-4, Acceptable Quality Level (AQL) of 2.5 percent defective for examination as specified in 4.4.3.2, 4.4.3.3 and 4.4.4.

4.4.3 Inspection procedures.

4.4.3.1 Tests. The quality conformance tests (lot acceptance tests) of the coating compound shall consist of all the tests specified in Table XI and such other tests as are necessary to check the manufacturer's data. If a test specimen fails to meet any of the tests required by this specification, the lot represented by the sample shall be rejected.

4.4.3.1.1 In addition, the Government reserves the right to have any of the tests called for by this specification, except weather resistance, rerun at a Government laboratory at any time within 6 months from the date of manufacture, as attested by the date appearing on the label. Samples for these tests shall be taken from a previously unopened container of the submitted acceptance sample, which shall have been stored as described in 4.6.1.2.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:38Z
Check the source to verify that this is the current version before use.

4.4.3.2 Examination of preparation for delivery. The samples selected in accordance with 4.4.2.3 shall be examined for packaging, packing, marking, leakage, and workmanship. Any package having one or more defects shall be rejected. If the number of defective packages exceeds the Acceptable Quality Level (AQL) of 2.5 percent defective in accordance with MIL-STD-105, the lot represented by the sample shall be rejected.

4.4.3.3 Examination for container fill. The size sample of 4.4.2.3 shall be adjusted to Inspection Level S-3 of MIL-STD-105 except that the sample size derived from MIL-STD-105 shall be rounded off to the next highest increment of five. At no time shall the sample size be less than five, unless the lot consists of less than five containers in which case all containers shall be used to determine the fill. The sample unit shall be one filled container. The lot shall be rejected if the weight of any container is less than that specified.

4.4.4 Examination of product. Conformance of the coating compound to the requirements for workmanship and any other requirements not covered by specific tests shall be determined by appropriate visual examination.

4.5 Test conditions. Unless otherwise specified herein, testing shall be performed at 73.4 $\pm$2°F (23 $\pm$1.1°C) and a relative humidity of 50 $\pm$4 percent.

4.5.1 Test panels. Except as otherwise specified herein, test panels shall be 6061 aluminum alloy conforming to QQ-A-250/11, condition T-6. The panels shall be 0.040 by 4 by 12 inches in size and treated with materials conforming to MIL-C-81706 to produce coatings conforming to MIL-C-5541. When specified, the cured coating system for test purposes shall include a topcoat over the coating compound as specified in 4.5.1.2.2.

4.5.1.1 Panel preparation. Epoxy-polyamide primer conforming to MIL-P-23377 shall be applied to the panels to a dry film thickness of 0.6 to 0.8 mils and then cured 1 hour at 160°F.

4.5.1.2 Application of coatings.

4.5.1.2.1 Coating compound. The coating compound shall be sprayed on the test panels of 4.5.1.1 in accordance with 4.6.4 to a dry film thickness of 0.070 $\pm$.010 inch. Coated panels shall be cured 4 hours at 120 $\pm$2°F prior to topcoating with polyurethane. Test panels not requiring topcoating shall be conditioned an additional 16 hours at 180 $\pm$2°F prior to conducting tests.

4.5.1.2.2 Polyurethane topcoat (MIL-C-81773, Color No. 17875). Test panels of 4.5.1.2.1 requiring polyurethane topcoat for tests specified herein shall be coated by conventional spray application. To the test panels of 4.5.1.2.1 which have been cured for 4 hours at 120 $\pm$2°F, apply the polyurethane topcoat to a dry film

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

thickness of 1 to 1.5 mils. Coated panels shall be conditioned for 16 hours at 180 ±2° F prior to conducting tests.

4.6 Test methods.

4.6.1 Component test methods. Components I, II, and III shall be tested in accordance with the applicable test method specified in Table XII, and the test methods specified in 4.6.1.1 through 4.6.2.

4.6.1.1 Condition in container. Components I, II, and III shall be stirred and observed for conformance to 3.5.4.

4.6.1.2 Storage stability. Each component container shall be placed in an area where they will not be agitated or disturbed for a period of 6 months at 65 ±25° F. After storage, the components shall be tested for conformance to the requirements of Tables II, III or IV as applicable.

4.6.2 Amine value.

4.6.2.1 Reagents.

a. Standard perchloric acid 0.1 N. Dilute 8.5 ml of 72 percent perchloric acid to 1 liter with reagent grade dioxane. 1/

b. Potassium Hydrogen phthalate, Primary Standard.

c. Methyl violet indicator solution, 0.2 percent. Weight/volume in chlorobenzene.

1/ CAUTION - 72 percent Perchloric Acid is a strong oxidant. Contact with other materials may cause fire or explosion, especially if heated. Causes severe burns.

PRECAUTIONS - Keep container closed and away from heat.
Store separately from, and avoid contact with, dehydrating agents and other materials.
Do not get in eyes, on skin, on clothing.
In case of spillage, flush with plenty of water and remove contaminated articles.
In case of contact, immediately remove all contaminated clothing and flush skin or eyes with plenty of water for at least 15 minutes; for eyes, get medical attention.

4.6.2.2 Procedure. For Class 1, Component III, weigh 0.3 ± .1 gms of sample to the nearest ± 0.1 mg into a 250 ml Erlenmeyer flask. Add 50 ml of glacial acetic acid and 3 drops of 0.2 percent methyl violet indicator. Swirl the flask gently until the sample is dissolved. Titrate with 0.1 N $HClO_4$ (in dioxane) to the first

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

TABLE XII

COMPONENT TESTS

COMPONENT CHARACTERISTIC	APPLICABLE METHOD IN FED. TEST METHOD STD. NO. 141	APPLICABLE ASTM TEST METHOD	REQUIREMENT TABLE
Viscosity			
Class 1			
Component I, No. 7 Spindle, 20 RPM	4287	D 2196	II
Component II, No. 3 Spindle, 20 RPM	4287	D 2196	III
Class 2			
Component I, No. 2 Spindle, 10 RPM	4287	D 2196	II
Component II, No. 2 Spindle, 50 RPM	4287	D 2196	III
Weight per gallon			
Component I	4184	D 1475	II
Component II	4184	D 1475	III
Volatile content			
Component I	4041	D 2369	II
Component II	4041	D 2369	III
Flash point			
Component I	4291	D 56	II
Component II	4291	D 56	III
Component III,			
Class 1		D 1310	IV
Class 2	4291	D 56	V

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

TABLE XII (Continued)

COMPONENT CHARACTERISTIC	APPLICABLE METHOD IN FED. TEST METHOD STD. NO. 141	APPLICABLE ASTM TEST METHOD	REQUIREMENT TABLE
Amine Value, Component III	See 4.6.2	See 4.6.2	IV (Class 1), V (Class 2)
Specific gravity, Component III	4183	D 891	IV (Class 1), V (Class 2)
Water content Component III	4082	D 1364	IV (Class 1), V (Class 2)
Infrared reflectance (Component III Class 1) 1/			

1/ When measured on a suitable spectrophotometer, the infrared reflectance shall conform to Figure 1.

color change from violet to blue. For Class 2, Component III, weight 0.6 ± .1 gms of sample using the same test procedure. Calculate the amine number and equivalent weight as follows:

CLASS 1, COMPONENT III

$$\text{Amine Number} = \frac{E \times 5.61}{F}$$

CLASS 2, COMPONENT III

$$\text{Amine Number} = \frac{E \times 5.61}{F \times 0.5}$$

Where E = milliliters of $HClO_4$ required for titration of sample

F = grams of sample used

4.6.3 Ingredient test methods.

4.6.3.1 Liquid polysulfide polymer.

4.6.3.1.1 Material. When measured on a suitable spectrophotometer, the infrared reflectance of the material shall conform to 3.6.1.

4.6.3.1.2 Properties. The physical properties of the polysulfide polymer shall be determined in accordance with Table XIII.

4.6.3.2 Ammonium salt of 4-nitroaniline-2-sulfonic acid. The properties of the ammonium salt shall be determined in accordance with NASA Specification No. A-16839(E).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:38Z
Check the source to verify that this is the current version before use.

TABLE XIII

POLYSULFIDE POLYMER PROPERTY TEST METHODS

PROPERTY	ASTM TEST METHOD
Color	ASTM D 1544
Moisture content 1/	ASTM D 1461
pH (Water extract) 2/	ASTM D 1417
Viscosity, No. 3 spindle	ASTM D 1638
Specific Gravity, Method A	ASTM D 1638

1/ Prepare the test sample by mixing 100 gm sample of liquid polymer with 100 gms of dry toluene.

2/ Sample preparation: Add 20 gm of dry toluene to 20 gms of liquid polymer and mix. Weigh 10 gms of this mixture into a 250 ml erlenmeyer flask, add 90 gms of water and shake 15 minutes, filter and determine the pH.

4.6.3.3 Toluene. The properties of the toluene shall be determined in accordance with TT-T-548.

4.6.3.4 Silica powder tests.

4.6.3.4.1 pH determination. Determine the pH of the silica powder in accordance with ASTM Test Method E70 with the exceptions listed in 4.6.3.4.1.1.

4.6.3.4.1.1 Apparatus. Use a Beckman Model G pH meter or equal with Beckman electrodes Nos. 39170 and 40485. As an alternate, Beckman Model 76 pH meter may be used with electrode Nos. 41236 and 40471. The stirrer shall be a Cenco Magnetic Stirrer, No. 18857 or equal with No. 21-5/8" Teflon Coated Stirring Bars No. 18854.

4.6.3.4.1.2 Procedure.

a. Weigh 4 grams of sample into a 250 ml beaker and add 100 ml of water (ASTM E 70) with a pH of 5.5 minimum.

b. Stir the silica powder water mixture vigorously for 5 minutes.

c. Stop the stirrer. When the liquid's agitation ceases, record the pH reading.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

4.6.3.4.2 Residue percent.

4.6.3.4.2.1 Apparatus.

a. No. 325 Sieve (RR-S-366).

b. Water line pressure regulator set for 16 psig.

c. Sprayco No. 5B, or equal, spray nozzle.

d. Funnel, Dimensions are: Mouth opening, 8.4 inches diameter; total height, 9 inches, stem length, 2.7 inches, stem diameter opening, 1.4 inches.

e. Washing hose.

4.6.3.4.2.2 Procedure. To one liter of water, disperse 10 grams of silica powder by hand. Pour the mixture into the funnel with the spray water set at 16 psig. Continue the spray for 45 seconds after the funnel is empty. Wash sides of funnel. After washing, the sieve containing the residue is dried for 20 minutes at 105 $\pm 2^{o}$C and weighed. Remove material, weight residue and reweigh sieve. The percent of residue shall be calculated as follows:

$$\text{Percent residue} = \frac{\text{Weight of residue}}{\text{Weight of sample}} \times 100$$

4.6.3.4.3 Bulk density.

4.6.3.4.3.1 Apparatus.

a. Cylinder container approximately 800 cc volume.

b. Balance, accurate to 0.1 gram.

4.6.3.4.3.2 Procedure. Slowly pour the powder into the container and allow to settle (care must be taken not to jar container). As the powder settles, more is added, keeping the powder heaped in the container. When settling has stopped (approximately 3-5 minutes), the excess is scraped off with a straight edge.

4.6.3.4.3.3 Calculations. Calculate the bulk density as follows:

a. Weight of container and material less weight of container = net weight of silica powder.

b. Bulk density (lbs/ft^3) = $\frac{\text{Net weight of silica powder}}{\text{Volume of container in cc}} \times 62.43$

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

4.6.3.4.4 Surface area.

4.6.3.4.4.1 Apparatus.

a. Beckman Model GpH Meter using Beckman External Fiber Type Calomel Electrode (Beckman No. 39170) with 30" lead and Beckman Expernal Type 42 Glass Electrode (Beckman No. 40485) or equal.

b. Cenco Magnetic Stirrer (No. 18851) with 21-5/8" Teflon Coated Stirring Bars (No. 18854) or equal.

c. 50 ml burette, capable of delivering quantities accurate to within 0.05 ml (i.e., Kemax No. 17060) or equal.

4.6.3.4.4.2 Reagents.

a. 0.1 N NaOH Solution.

b. 0.1 N HCl Solution.

c. 20 percent NaCl Solution (W/W).

4.6.3.4.4.3 Procedure. Transfer 2.5 grams of silica powder into a 400 ml beaker and add 250 ml of 20 percent NaCl solution. Place suspension on stirrer at maximum rate and adjust pH to 4.0 by addition of HCl or NaOH. Titrate to pH of 9.0 by addition of 0.1 N NaOH at a rate of approximately 1 drop/second. Allow to stir for 5 more minutes while holding the pH at 9.0 by addition of NaOH in 1 drop increments. Calculate the surface area as follows:

$$\text{Surface area} = (\text{ml} - \text{ml Blank}) \times N\,(\text{NaOH}) \times 138.6$$

NOTE

A correlation of surface area by nitrogen adsorption (Brunauer-Emmet-Teller Method) versus volume of titrant used has been established and is now used as a basis for calculating surface area).

4.6.3.5 Liquid epoxy resin. When measured on a suitable spectrophotometer, the infrared reflectance of the epoxy resin shall conform to 3.6.5. The physical properties shall be determined in accordance with Table XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

TABLE XIV

LIQUID EPOXY RESIN PROPERTY TESTS

PROPERTY	ASTM TEST METHOD
Epoxide equivalent	ASTM D 1652
Viscosity	ASTM D 445
Color	ASTM D 1544
Refractive Index	ASTM D 1747
Flash Point	ASTM D 1310
Density	ASTM D 1217

4.6.3.6 Glass and silica fiber tests.

4.6.3.6.1 Bulk density. Determine the bulk density as specified in 4.6.3.4.3 except that the calculations shall be as follows:

$$\text{Bulk density gms/CC} = \frac{\text{Net weight of fibers in grams}}{\text{Volume of container in CC}}$$

4.6.3.6.2 Ignition loss. Determine the ignition loss in accordance with Method 4462 of Fed. Test Method Std. No. 141 except as follows:

Sample size shall be approximately 25 gms. For glass fibers, the sample shall be subjected to 1200° F in the muffle furnace for 25 to 27 minutes. For silica fibers, the sample shall be subjected to 3200° F in the muffle furnace for 25 to 27 minutes.

4.6.3.6.3 Fiber length and diameter. The diameter and length of the fibers shall be determined in accordance with ASTM Test Method E 20. A laboratory microscope with 10X and 43X objectives and a filar measuring eyepiece or microprojector may be used. For contrast of the fiber, anisole placed on the edge of the cover glass will readily flow under the coverglass to the fibers.

4.6.3.7 Methyl Isobutyl Ketone. The properties of the methyl isobutyl ketone shall be determined in accordance with TT-M-268.

4.6.3.8 Methyl Ethyl Ketone. The properties of the methyl ethyl ketone shall be determined in accordance with TT-M-261.

4.6.4 Application. The three components of the coating compound shall be admixed as specified in 3.2.1.

4.6.4.1 Equipment. The admixed coating compound shall be applied to panel surfaces with the following equipment.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:38Z
Check the source to verify that this is the current version before use.

a. Binks airless spray gun, Model No. 50 or equal.

b. Tip No. 54-2738.

c. Tank No. C-255, Serial No. 54-2736.

d. Air intake pressure - 100 psi.

e. Head pressure - 65 - 70 psi.

4.6.4.2 Procedure.

4.6.4.2.1 Class 1 and 2 compounds. The spray gun shall be held 8-12 inches from the panel surface while spraying. For Class 1 compound, a total of 4 passes shall be made with a 30 minute interval between each pass. For Class 2 compound, a total of 7 passes shall be made with a 30-minute interval between each pass. Total dry film thickness after the required passes shall be 70 $\pm$10 mils. Air-dry the coating at ambient conditions for a minimum of 72 hours before testing. The viscosity of the admixed material may be adjusted by thinning with methyl ethyl ketone. The thinner may be added in the quantity of up to approximately 5 percent by volume of the admixed components.

4.6.5 Thermal efficiency.

4.6.5.1 Panel preparation. Prepare eight panels each 0.040 by 8 by 8 inches in size and eight discs each 1-15/16 inch in diameter by 0.040 inch in thickness from aluminum (6061) sheets conforming to QQ-A-250/11, Condition T-6. Cut a 2-inch hole in the center of each panel and cement the discs in the holes with a metal-to-metal type epoxy bonding compound capable of withstanding high temperatures. Cementing the discs in the panels reduces the thermal conductance effects from the edges. The test panels shall be prepared as specified in 4.5.1, 4.5.1.1 and coated with the compound in accordance with 4.6.4 and subsequently machined to a uniform film thickness of 70 $\pm$5 mils. After machining, spray the polyurethane top coat over the compound as specified in 4.5.1.2.2. Save three panels for the humidity resistance test 4.6.11, and two panels for the weather resistance test (4.6.13).

4.6.5.2 Procedure. This test exposes the coating to a fire environment where a total flux of 10 BTU/Ft^2 - sec as measured by an asymptotic calorimeter is 90 percent radiative in slightly fuel rich condition. The average emissivity of the source of the radiant heat flux shall be at least equal of 0.9. Aviation fuel (JP-4 or JP-5) conforming to MIL-T-5624 shall be used to fire the unit. The coated panel thickness shall be measured and recorded. When ready to test, the panel shall be inserted in a transite mask or any thermally insulated holder which allows the panel to rest on about 1/16 to 1/8 inch of the edges of the coated surface. The backside of the test panel shall be insulated with nominally 6 lbs/ft^3 refractory fiber insulation.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:38Z
Check the source to verify that this is the currentversion before use.

At time, T = O, the masked panel shall be placed over the fire box and a thermocouple in contact with the back of the disc records the temperature rise on the panel. The maximum initial panel temperature prior to fire testing shall be 90° F. The flux shall be uniform over the entire panel and the center disc shall be approximately 8-10 percent of the exposed area and shall be representative of a larger semi-infinite panel. The time-temperature data and the film thickness in mils shall be recorded and a report made of the time, in seconds, to reach the temperature of 260° C (500° F), for the recorded mil thickness of the coating. The test results shall conform to 3.8 and the data reported shall be an average of three determinations.

4.6.6 Tensile properties. The tensile properties shall be determined in accordance with Method 1011 of Fed. Test Method Std. No. 406 with the following exceptions: The film of coating shall be applied to a dry film thickness of 70 ± 10 mils as specified in 4.5.1 and 4.5.1.2.1 and cured 24 hours at 158° F plus a minimum of 24 hours at 75° F before test. The crosshead speed of the tensile machine shall be 0.2 to 0.5 inches/minute. The film may be cast on thin aluminum sheeting treated with silicone mold release for easier removal.

4.6.7 Coating properties.

4.6.7.1 Color. The coated panels of 4.6.8 shall be visually examined for conformance to 3.10, prior to their immersion in the lubricating oil.

4.6.7.2 Hardness. Components I, II, and III shall be admixed in the ratio specified in 3.2.1 and sprayed to a thickness of 70 ± 10 mils on panels prepared as specified in 4.5.1 through 4.5.1.2.1. The coated panels shall be cured at room temperature for 72 hours and then tested for Shore A hardness in accordance with Method 1082 of Fed. Test Method Std. No. 406 for conformance to 3.11.

4.6.7.3 Adhesion. Panels shall be aluminum alloy conforming to QQ-A-250/5, T - O Condition, and anodized in accordance with MIL-A-8625, Type II. They shall be 0.020 by 5 by 12 inches in size and finished with the primer and coating compounds specified in 4.5.1.1, 4.5.1.2.1, and 4.5.1.2.2. The panels shall be tested in accordance with Method 6221 of Fed. Test Method Std. No. 141 and checked for conformance to 3.12.

4.6.8 Lubricating oil resistance. Panels shall be prepared as specified in 4.5.1 through 4.5.1.2.1. The 70 ± 10 mil films shall be cured 24 hours at 75° F, plus 24 hours at 158° F, plus a minimum of 24 hours at 75° F before oil immersion. The panels shall then be immersed in lubricating oil composed of 95 percent of di-2 ethyl-hexyl sebacate 1/, and 5 percent tricresyl phosphate, by weight, at a temperature of 250 ± 4° F for a period of 4 hours. Four hours after removal, the panels shall be examined for conformance to 3.13.1.

1/ di-2 ethyl hexyl sebacate may be obtained commercially as "Plexol 201" from Rohm and Haas.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

4.6.9 Fungus resistance. Panels shall be prepared as specified in 4.5.1 through 4.5.1.2.2 and tested for resistance to fungus in accordance with Method 508, Procedure I of MIL-STD-810. Upon completion of the fungus test, check for conformance to 3.13.2.

4.6.10 Impact resistance.

4.6.10.1 Panel preparation. Panels shall be aluminum alloy conforming to QQ-A-250/5, T - O Condition and anodized in accordance with MIL-A-8625, Type II. They shall be 0.020 by 5 by 12 inches in size, and finished as follows: Apply epoxy primer as specified in 4.5.1.1. Over the epoxy film, apply the coating compound to a thickness of 70 ± 10 mils. The coated panels shall be cured at room temperature for 72 hours before conducting tests. After cure, the compound shall be topcoated with polyurethane as specified in 4.5.1.2.2.

4.6.10.2 Impact test. Place one prepared panel from 4.6.10.1, film side up over the hole of the cylindrical base of a Gardner Heavy Duty Impact Tester 1G-1120. Determine the maximum impact resistance in inch-pounds at which no failure occurs. Repeat the test on the reverse side of another panel from 4.6.10.1 with the film side down. Check for conformance to 3.13.3.

4.6.11 Humidity resistance. Three panels prepared as specified in 4.6.5.1 shall be subjected to 336 hours of exposure at 100 percent relative humidity at 120° F in accordance with the method specified in ASTM D 1748.

4.6.12 Resistance to temperature shock. Three panels prepared as specified in 4.5.1 through 4.5.1.2.2 shall be tested in accordance with Method 503 of MIL-STD-810B except as follows:

a. The internal hot chamber shall be maintained at 140° F and 95 percent RH.

b. The internal cold chamber temperature shall be -40° F.

c. Panels shall be cycled every 24 hours in lieu of 4.

d. The complete test period shall be 28 days.

4.6.13 Weather resistance. Two panels shall be prepared as specified in 4.6.5.1 and exposed outdoors for one year in the vicinity of Miami, Florida in accordance with ASTM Test Method D 1014. The panels shall be mounted at 45 degrees facing South. After exposure, the panels shall be checked for conformance to 3.13.6 and subjected to the thermal efficiency test (4.6.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Component unit packages. Component unit packages shall be furnished as a kit, consisting of 3 separately packaged components. Each component (see 3.2.1) shall be furnished in containers specified in 5.1.1.1.1 through 5.1.1.1.3. The quantity of each component and size of container shall be as specified by the procuring activity (see 6.2). Interior surface containers shall be treated as required to preclude content contamination.

5.1.1.1.1 Cans. Cans, one gallon and smaller, shall conform to TT-P-143.

5.1.1.1.2 Pails. Pails shall conform to PPP-P-704, Type II, Class 3; or Type III, Class 3 or 4.

5.1.1.1.3 Drums. Drums shall conform to PPP-D-729, Type III or IV.

5.1.2 Level C. The coating compound components shall be packaged in unit capacity containers to form a kit in accordance with the supplier's commercial practice.

5.2 Packing. Packing shall be Level A or C as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Cans, pails and drums. Unless otherwise specified (see 6.2) component cans, pails and drums, shall be packed as a kit in accordance with Level A requirements of TT-P-143.

5.2.2 Level C. Material packaged as specified in 5.1.2 shall be packed in such a manner that will insure acceptance by common carrier at lowest rates and safe delivery at destination. Shipping containers shall comply to the Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. Marking for shipment shall be in accordance with MIL-STD-129 and shall include the following nomenclature and information:

Class (1 or 2) as applicable
Component Identification: (Component I, II or III)
as applicable
Manufacturer's Name or Trademark

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

Date Manufactured, Month and Year
Lot Number and Batch Number
Net Contents
MIL-C-81947A
Flash Point (In accordance with ICC Regulations)

5.3.1 Additional markings. Individual containers of each kit furnished shall be marked with the following labels:

"CAUTION

May Cause Skin & Eye Irritation!

DO NOT GET IN EYES, AND AVOID PROLONGED OR REPEATED CONTACT WITH SKIN.

In case of contact immediately flush skin or eyes with plenty of water for at least 15 minutes; for eyes, get prompt medical attention. Wash clothing before reuse."

"CAUTION

This is one component of a three component coating which WILL NOT HARDEN unless all components are mixed together."

"INSTRUCTIONS FOR USE

(Class 1 or 2, as applicable)

CLASS 1

Stir each component thoroughly. Add Component III to Component I and stir thoroughly.
Add Component II to the admixture of Components I and III in the ratios listed below and stir thoroughly.

WEIGHT RATIO

COMPONENT	PARTS BY WEIGHT
I	841.0
II	117.5
III	10.0

CLASS 2

Stir each component thoroughly. Add Component III to Component I and stir thoroughly.
Add Component II to the admixture of Components I and III in the ratios listed below and stir thoroughly.

WEIGHT RATIO

COMPONENT	PARTS BY WEIGHT
I	761.7
II	133.6
III	39.2"

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

6. NOTES

6.1 Intended use. The thermal insulating coating compound is intended for use as an exterior protective coating on aircraft ordnance items. The coatings are formulated to meet applicable environmental conditions and also to delay an explosive reaction when the coated ordnance item is exposed to high flux thermal environments. Usage of both Class 1 and 2 coating compound requires a topcoat of polyurethane in the specified color. For design purposes, users should compare the differences in physical properties between Class 1 and 2 compounds.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Components (see 3.5.1, 3.5.2 and 3.5.3).

c. Level of packaging and packing (see Section 5).

d. Components packaged and packed as a kit (see Section 5).

e. Quantity in pounds (avoirdupois).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Requests for information pertaining to qualification of products covered by this specification may be obtained from the Commanding Officer, U. S. Naval Air Development Center, Warminster, Pennsylvania 18974, Attention: Code 30222.

6.4 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

6.5 Ingredients. The ingredients of Table I which when properly processed, have produced a coating compound meeting the requirements of this specification. The following list of approved proprietary raw materials is not to be construed as an endorsement thereof or as precluding similar materials from other proprietary sources. Such products may prove equivalent or even superior in performance to the ones listed.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:38Z
Check the source to verify that this is the current version before use.

	INGREDIENT (Table I)	SOURCE
a.	Polysulfide polymer, liquid	LP-3, Thiokol Chemical Corporation
b.	Ammonium salt of 4-nitro-aniline-2-sulfonic acid	Nyanza, Inc.
c.	Epoxy resin	Epon 828, Shell Chemical Co.
d.	Silica powder	CAB-O-Sil(M-5), Cabot Corp.
e.	Fiberglass fibers	701A-K-1/32, Owen Corning Fiberglas Corp.
f.	2, 4, 6-Tri (dimethyl-aminomethyl) phenol	DMP-30, Rohm and Haas Company
g.	Toluene	TT-T-548
h.	Methyl Isobutyl Ketone	TT-M-268
i.	Methyl ethyl Ketone	TT-M-261
j.	Silica fibers	Refrasil, Hitco, Materials Division

6.6 Class 1 coating compound may be referred to as "Naval Air Development Center formulation P-124" and Class 2 coating compound as "NASA - Ames Research Center Formulation 313."

Custodian:
Army - MR
Navy - AS
Air Force - 84

Review Activity:
Army - AV

User Activity:
Navy - OS

Preparing Activity:
Navy - AS
(Project No. 8010-0777)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL	OMB Approval No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE: MIL-C-81947A COATING COMPOUND, THERMAL INSULATION, THREE-COMPONENT, INTUMESCENT

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO (If "Yes", in what way?)

4. REMARKS

SUBMITTED BY (Printed or typed name and address - Optional)	TELEPHONE NO.
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:38Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Air Engineering Center
Lakehurst, NJ 08733

OFFICIAL BUSINESS

POSTAGE AND FEES PAID
DEPARTMENT OF THE NAVY
DOD 316



Engineering Specifications and Standards Department (Code 93)
Naval Air Engineering Center
Lakehurst, NJ 08733

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:38Z
Check the source to verify that this is the current version before use.